

Mathematical Description of I-O Based GDP Impact Calculation

1. Data Preparation

For each country c and year t , we load:

- The input-output matrix $II^{(c,t)}$.
- Final household consumption expenditure $HFCE^{(c,t)}$.
- Gross Domestic Product vector: $GDP^{(c,t)} = VALU^{(c,t)}$.
- Total output vector: $output^{(c,t)} = OUTPUT^{(c,t)}$.

2. Leontief Inverse Matrices

Open Model

We define the technical coefficients matrix:

$$T = \frac{II}{\text{output}}$$

Then the Leontief inverse is:

$$L = (I - T)^{-1}$$

Closed Model

The closed model appends:

- A new column to II : final demand from household income ($HFCE$).
- A new row: employee compensation vector (E).

Denoting the extended matrix by II_c and extended output by $output_c$, the new matrix of technical coefficients is:

$$T_c = \frac{II_c}{\text{output}_c}$$

The corresponding Leontief inverse is:

$$L_c = (I - T_c)^{-1}$$

3. Multipliers

Output Multipliers

$$m_o = L \cdot \mathbf{1}$$
$$m_{oc} = L_c^{\text{truncated}} \cdot \mathbf{1}$$

Income Multipliers

Let E_j be employee compensation, x_j be output:

$$\frac{E_j}{x_j}$$

Income multiplier matrix:

$$M_h = L \cdot \text{diag} \left(\frac{E_j}{x_j} \right) \quad \text{and} \quad M_{hc} = L_c \cdot \text{diag} \left(\frac{E_j}{x_j} \right)$$

GDP Multipliers

Let G_j be GDP value-added:

$$\frac{G_j}{x_j}$$

$$M_g = L \cdot \text{diag} \left(\frac{G_j}{x_j} \right) \quad \text{and} \quad M_{gc} = L_c \cdot \text{diag} \left(\frac{G_j}{x_j} \right)$$

4. Decomposing Effects

Let n be the number of sectors.

- **Direct:** Identity matrix or diagonal of $\frac{E_j}{x_j}$ or $\frac{G_j}{x_j}$.

- **Indirect:**

$$\text{indirect}_o = L - I, \quad \text{indirect}_h = M_h - \text{diag} \left(\frac{E_j}{x_j} \right)$$

- **Induced:**

$$\text{induced}_o = L_c^{\text{trunc}} - L, \quad \text{induced}_h = M_{hc}^{\text{trunc}} - M_h$$

5. Final Demand and Impacts

Let f be the final demand vector (sum of final demand columns):

$$\text{Direct Impact} = \text{direct multiplier matrix} \cdot f$$

$$\text{Indirect Impact} = \text{indirect multiplier matrix} \cdot f$$

$$\text{Induced Impact} = \text{induced multiplier matrix} \cdot f$$

$$\text{Total Impact} = L_c^{\text{trunc}} \cdot f$$

6. GDP Impact Table

Construct a table of the form:

$$\text{GDP Impact}_{ij}^{(\text{type})}, \quad \text{type} \in \{\text{direct, indirect, induced, total}\}$$

with i as selling sector and j as buying sector.